PAPER – 4: COST AND MANAGEMENT ACCOUNTING

QUESTIONS

PART I - Case Scenario based MCQs

Integrated

1. Mr. Linde is a German national, who came to India again on 1st April, 2024. He represents his company and wants to start business in India as well. His company expertise in the manufacturing of Industrial machines. Recently launched "Make in India" movement has motivated Mr. Linde thinks that this might be the perfect opportunity for his company to establish his company in India.

   Last, Mr. Linde came to India on 1st April, 2012. He purchased a land for ₹ 50,00,000 and constructed a building by spending ₹ 16,00,000. After that he opened a Private limited company in that building. He spent another ₹ 2,80,000 for this. He also employed 3 people for survey and to understand the need of Indian customers and spent ₹ 1,50,000 in salaries.

   He was disappointed in the response of market, who were importing everything from China back then. He closed the office & went back to Germany. All these years the office was closed and only an amount of ₹ 12,500 per month was paid to a guard and property tax was also paid. Property tax was paid on an average of ₹ 18,000 per year.

   Now when Mr. Linde is back, he opens the office and starts to plan on how this time he will capture the Indian market.

   Expenses started to incur as soon as the office opened:

- Salaries of staff ₹ 2,50,000 per month.
- Electricity, water, & maintenance of office at ₹ 50,000 per month.
- Security staff at ₹ 15,000 per month.

Linde plans to purchase a land in Manesar which will be used for the factory. After a search he found an appropriate land and purchased a land for ₹ 1.50 crores. He handed over the land to a SPV company of a REIT to build a state of the art facility for their factory. Factory will be built in 2 years. They will spend ₹ 85 lacs each year for this construction.

Linde, back in the Noida office, made 3 departments:

(1) Office and administration

(2) Sales and marketing

(3) Account and Finance

Expenses for these departments (except for salaries) are expected to be:-

- Office and administration = ₹ 95,000 per year
- Sales and marketing = ₹ 1,12,000 per year
- Accounts and Finance = ₹ 88,000 per year

Office overheads are to be bifurcated in these departments on the basis of their individual spending ratio.

Technology is developed in Germany but at present its execution is not required. Therefore, they do not require any expert as of now and also because the factory is not ready.

Mr. Linde, being the only person representing his company and lone German in the Indian office feels difficult to manage everything as he finds Indian corporate environment very challenging. He asked his company to deploy another German manager to India. This will cost the company additional two million Indian national rupee per year to relocate this additional manager in India. The German management is divided on this decision. The ones who disagree say "Mr. Linde is competent enough to run a small extension of our company in India. We will allocate more resources to Indian subsidiary when actual operations will start, till then everything can be managed by Mr. Linde alone. Right

Indian Company is itself a cost centre and we are already paying him 3.5 million INR annually, therefore we are not ready to invest until it starts generating revenue".

Linde has another opportunity to relocate the head office, also, to Manesar, where the factory building is being constructed. The distance between head office and factory will reduce greatly, which will be highly beneficial when the factory will become operative. He will have to sale the old office in Noida, which will be sold at ₹ 2.50 crores and purchase a ready-made building in Manesar for ₹ 3.75 crores. This new building will have larger space that can accommodate the future needs for space, when company will grow. It seems to be like perfect investment opportunity to Linde.

Expenses in this new building are expected to be:-

- Salaries of staff ₹ 3,00,000 per month
- Electricity, water, & maintenance of office at ₹ 80,000 per month.
- Security at ₹ 30,000 per month.

Indexed cost of building in Noida is ₹ 2.25 crores and tax on long-term capital gain is 12.5%.

On the basis of above information, answer the following 5 MCQs:

i. Find out an avoidable cost till the factory becomes operative. What is its value?

(a) 20,00,000

(b) 49,20,000

(c) 98,40,000

(d) 40,00,000

ii. Find out the total of Sunk and shut down cost in the given case study. Select the correct option from below.

(a) 4,30,000

(b) 70,30,000

(c) 24,46,000

(d) 90,46,000

iii. What is total out-of-pocket cost for the company in Noida branch, after factory land in Manesar is purchased, till the factory operation begins?

(a) 3,21,50,000

(b) 1,51,50,000

(c) 81,50,000

(d) 40,75,000

iv. What will be out of pocket expenses incurred in relocation of Head office to Manesar?

(a) 3,75,00,000

(b) 1,28,12,500

(c) 1,25,00,000

(d) 4,24,20,000

v. How much is the unexpired cost of the Noida office as on 1st October, 2024, if salaries to all the employees are paid till 31st March, 2025?

(a) 33,40,000

(b) 30,00,000

(c) 15,90,000

(d) 15,00,000

Activity based Costing

2. The HomeMart is the latest trending brand offering home improvement appliances with broadest selection of products with highly competitive prices. The sale is increasing year by year with huge multiples. Current year also the sales reached triple the last year. The reason being company having good customer support where it provides after sales assistance over phone per item sold. Though it costs only Re. 1 per item sold to the company, it enhanced to ₹ 49,15,200 last year making a huge impact on the total support cost.

All the company's appliances have been majorly categorised into three product lines namely Fancy fans, Home decors, Assembled furniture. During the current year, the company's revenue as generated is ₹ 3,80,88,000, ₹ 10,08,28,800 and ₹ 5,80,75,200 respectively. However, the cost of god sold is ₹ 2,88,00,000, ₹ 7,20,00,000 and ₹ 4,32,00,000 respectively.

In business, there's a saying "The packaging sells the product the first time, but what's inside sells the product a second time". Following the saying, the company has the policy of taking back the cartons of the products sold relating to Fancy fans to reduce the packaging cost. However, for smooth returning of cartons, the company has to incur certain carrier cost on its own which is ₹ 5,76,000 for the current year and allocating the same directly to the said product.

Some other information relating to each of the product lines is provided below:

	Fancy fans	**Home decors**	**Assembled furniture**
Items sold	12,09,600	1,05,98,400	29,37,600
Number of deliveries received	600	4,380	1,320
Number of purchase orders placed	720	1,680	720
Hours of shelf-stocking time	1,080	10,800	5,400

The company also provides the following basis of cost allocation:

Activity	**Description of activity**	**Total Cost**	**Cost-allocation base**
Delivery	Physical delivery and receipt of products	1,20,96,000	6,300 deliveries
Ordering	Placing of orders for purchases	74,88,000	3,120 purchase orders

Shelf stocking	Stocking of products in warehouse	82,94,400	17,280 hours of shelf-stocking time

The company wants you to FIGURE OUT the following to ascertain which of the product line is more profitable:

i. The total support cost and its percentage to the cost of goods sold would be:

(a) ₹ 3,33,69,600 and 23.17%

(b) ₹ 4,32,00,000 and 30%

(c) ₹ 3,33,69,600 and 30%

(d) ₹ 4,32,00,000 and 23.17%

ii. Operating income as a percentage of revenues of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated on the basis of cost of goods sold would be:

A. 6.87%, 12.05% and 8.38% respectively

B. 12.05%, 6.87% and 8.38% respectively

C. 1.70%, 7.17% and 3.30% respectively

D. 7.17%, 3.30% and 1.70% respectively

iii. The cost driver rate relating to Delivery, Ordering, Shelf stocking and Customer support would be:

(a) Delivery- ₹ 1,920 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- Re. 1 per item sold

(b) Delivery- ₹ 2,400 per delivery, Ordering- ₹ 1,920 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- Re. 1 per item sold

(c) Delivery- ₹ 1,920 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- ₹ 3 per item sold

(d) Delivery- ₹ 480 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹1,920 per stocking hour and Customer support- ₹ 3 per item sold

iv. Operating income of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated using an activity-based costing system would be:

(a) ₹ 16,84,800, ₹ -2,05,92,000 and ₹ -7,92,000 respectively

(b) ₹ 3,64,03,200, ₹ 12,14,20,800 and ₹ 5,88,67,200 respectively

(c) ₹ 92,88,000, ₹ 2,88,28,800 and ₹ 1,48,75,200 respectively

(d) ₹ 41,04,000, ₹ 6,04,800 and ₹ 50,83,200 respectively

v. Operating income as a percentage of revenues of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated using an activity-based costing system would be:

(a) 4.42%, -20.42% and -1.36% respectively

(b) 10.78%, 0.60% and 8.75% respectively

(c) 24.39%, 28.59% and 25.61% respectively

(d) 4.39%, 8.59% and -1.36% respectively

Employee Cost and Direct Expenses

3. Mr. A works in a manufacturing company where he is paid bonus according to the Halsey 50% plan, besides the normal wages. The relevant data is as below:

Time Rate (per hour)	₹ 100
Time allowed	10 hours
Time taken	5 hours
Time saved	5 hours

Mr. A believes that his bonus under Halsey system is getting reduced by 50%, thus intending to shift towards Rowan Premium Plan.

You are required to CALCULATE the total earnings of Mr. A as per Halsey plan and Rowan Premium plan & ENUMERATE the reason for difference in both the earnings.

(a) Total earnings as per Halsey plan- ₹ 500 and as per Rowan Premium plan- ₹ 750. Earnings under Halsey Plan is lower than that of Rowan Premium plan as the bonus is getting reduced by 50%.

(b) Total earnings as per Halsey plan- ₹ 750 and as per Rowan Premium plan- ₹ 500. Earnings under Rowan Premium plan is lower than that of Halsey plan as the actual time taken is 50% of the time allowed.

(c) Total earnings as per Halsey plan- ₹ 750 and as per Rowan Premium plan- ₹ 750. When the actual time taken is 50% of the time allowed, the earnings under Halsey and Rowan Plans are equal.

(d) Total earnings as per Halsey plan- ₹ 250 and as per Rowan Premium plan- ₹ 250. Total earnings under both the Plans are equal as the time taken under both the plans are same.

Cost Accounting Systems

4. WHICH of the following is the correct journal entry as would appear in the cost books when Material (Direct) is issued to production?

(a)	Store Ledger Control A/c	Dr.	xxx	
	To Work-in-Process Control A/c			xxx
(b)	Store Ledger Control A/c	Dr.	xxx	
	To Production Overhead Control A/c			xxx
(c)	Production Overhead Control A/c	Dr.	xxx	
	To Store Ledger Control A/c			xxx
(d)	Work-in-Process Control A/c	Dr.	xxx	
	To Store Ledger Control A/c			xxx

Process & Operation Costing

5. A product passes through two processes. The output of Process-I is treated as the raw material of Process-II. The cost incurred at Process- II is as follows:

Particulars	Process-II (₹)
Transferred from Process-I A/c	27,85,700
Materials issued	10,00,000
Labour	2,00,000
Manufacturing overhead	5,00,000

The output of each process is as under:

Process	Output	Normal Loss
Process-I	48,750 units	2%
Process-II	47,000 units	5%

No stock of materials or of work-in-process was left at the end.

You are required to CALCULATE the value of Abnormal Gain/Loss in Process-II A/c.

(a) Abnormal Gain of ₹ 66,626

(b) Abnormal Loss of ₹ 66,626

(c) Abnormal Gain of ₹ 72,776

(d) Abnormal Loss of ₹ 72,776

Joint Products and By products

6. Sterling Industries manages various manufacturing processes. In process I, joint products P1 and P2 are produced in the ratio of 6:4 in units from the raw material input. A normal loss of 2% of the raw material input is expected in this process, with losses having a realizable value of ₹ 12.5 per kg. The company has no work in progress. The joint costs are apportioned between the joint products using the physical measure basis.

The following information relates to process I for last month:

Raw materials input	75,000 kg (at a cost of ₹ 4,76,250)
Direct labour	₹ 2,25,000
Direct expenses	₹ 67,500
Production Overheads	110% of direct labour cost
Abnormal gain	1,250 kg

You are required to CALCULATE the number of unit and its value of the Normal loss, Abnormal gain and joint products P1 & P2.

(a) Normal loss: 1,500 units ₹ 16,964; Abnormal gain: 1,250 units ₹ 18,750; Product P1: 44,850 units ₹ 6,08,678 and Product P2: 29,900 units ₹ 4,05,786.

(b) Normal loss: 1,250 units ₹ 18,750; Abnormal gain: 1,500 units ₹ 16,964; Product P1: 44,850 units ₹ 6,08,678 and Product P2: 29,900 units ₹ 4,05,786.

(c) Normal loss: 1,500 units ₹ 18,750; Abnormal gain: 1,250 units ₹ 16,964; Product P1: 44,850 units ₹ 6,08,678 and Product P2: 29,900 units ₹ 4,05,786.

(d) Normal loss: 1,500 units ₹ 18,750; Abnormal gain: 1,250 units ₹ 16,964; Product P1: 44,850 units ₹ 4,05,786 and Product P2: 29,900 units ₹ 6,08,678.

Marginal Costing

7. PR Ltd. sells two types of pen, ball pen and gel pen. Currently, the company is expecting to sell 6,000 units of ball pen along with 3,600 units of gel pen in the coming month. Other information as forecasted is provided below:

Particulars	Ball pen	Gel pen
Selling price (per unit)	₹ 150	₹ 100
Variable cost (per unit)	₹ 90	₹ 60
Contribution (per unit)	₹ 60	₹ 40
Fixed Costs	₹ 3,36,000	

You are required to CALCULATE the Composite Break-even Batch and individual break-even of the pens (in units).

(a) Composite Break-even Batch- 6,400 batches, Break-even units of Ball pen- 4,000 units, Break-even units of Gel pen- 2,400 units.

(b) Composite Break-even Batch- 800 batches, Break-even units of Ball pen- 500 units, Break-even units of Gel pen- 300 units.

(c) Composite Break-even Batch- 800 batches, Break-even units of Ball pen- 4,000 units, Break-even units of Gel pen- 2,400 units.

(d) Composite Break-even Batch- 6,400 batches, Break-even units of Ball pen- 500 units, Break-even units of Gel pen- 300 units.

PART-II Descriptive Questions

Material Cost

8. Catalist Ltd. is a distributor of industrial chemicals, providing the chemical in drum packaging.

Each drum of the chemical costs ₹ 200 from a supplier and the company sells it for ₹ 240.

Annual demand is estimated to be for 2,50,000 drums.

The cost of delivery is estimated at ₹ 100 per order and the annual variable holding cost per drum at ₹ 180 plus 10% of purchase cost.

Based on above data, the managing director calculates the economic order quantity and suggests that it should serve as the foundation for purchasing decisions in upcoming periods.

However, the purchasing manager states that the managing director has ignored his share of bonus of 10% of the amount by which total annual inventory holding and order costs (before such remuneration) are below ₹ 2,00,000. He further points out that the suppliers also offer quantity discounts on purchase orders, i.e. if the order size is 1,000 drums or above, the price per drum is only ₹ 199.60, compared to ₹ 200 when an order is between 500 and 999 drums.

You are required to:

(i) CALCULATE the economic order quantity as calculated by the company's managing director.

(ii) COMMENT whether Catalist Ltd. can look forward to the quantity discount offered for purchasing 1,000 drums, after CALCULATING total cost considering purchasing manager's bonus along with supplier quantity discounts.

Employee Cost and Direct Expenses

9. Pi and Qu are part of a manufacturing team that handles multiple jobs within the production process. Detailed description of their respective monthly wages is provided below:

The standard working hours for the month are 210. Overtime is compensated at twice the sum of basic wages and dearness allowance. The employer's contributions to State Insurance and the Provident Fund are equal to the employees' contributions. Employees Pi and Qu were assigned to jobs A, S, and D in the following proportions:

Particulars		Pi	Qu
(i)	Basic Wages (₹)	12,000	15,000
(ii)	Dearness Allowance	50%	50%
(iii)	Contribution to provident Fund (on basic wages)	10%	10%
(iv)	Contribution to Employee's State Insurance (on basic wages)	1.75%	1.75%
(v)	Overtime (Hours)	8	--

Jobs	A	S	D
Worker Pi	30%	30%	40%
Worker Qu	30%	50%	20%

Overtime was done on job S.

You are required to CALCULATE the earnings of Pi and Qu and allocate the employee cost to each job A, S, and D.

Overheads- Absorption Costing Method

10. X Corp. produces two products, X and Y. The production division is made up of two manufacturing departments, P1 and P2, along with two support departments, S1 and S2.

Pre-determined Overhead Rates are applied in the production departments to allocate factory overhead costs to the products. The rate for Department P1 is determined by direct machine hours, while Department P2's rate is based on direct labor hours.

A technical assessment of the apportionment of expenses of service departments is as under:

	Production Department	
	P_1	**P_2**
Service Dept. 'S_1' (ratio)	3	3
Service Dept. 'S_2' (ratio)	30	15

Following budgeted data is available:

Factory overheads for the year:

Production Departments		**Service Departments**	
P_1	**P_2**	**S_1**	**S_2**
₹ 2,80,50,000	₹ 2,39,25,000	₹ 66,00,000	₹ 49,50,000

Details relating to production of the Products X and Y is as follows:

	Products	
	X	**Y**
Budgeted output (units)	1,00,000	60,000
Budgeted raw-material cost per unit (All materials are used in Department P_1 only.)	₹ 660	₹ 825

Budgeted time required for production per unit	P_1: 1.5 machine hours P_2: 2 Direct labour hours	P_1: 1.0 machine hours P_2: 2.5 Direct labour hours
Average wage rate budgeted in Department P_2	₹ 396 per hour	₹ 412.50 per hour

Following actual data is available (for the month of December, 2024):

	Products	
	X	**Y**
Actual output (units)	8,000	6,000
Actual hours worked for production	P_1: 12,200 machine hours P_2: 16,400 Direct labour hours	P_1: 8,300 machine hours P_2: 14,800 Direct labour hours
Raw materials cost	₹ 53,79,000	₹ 50,16,000
Wages paid	₹ 65,10,900	₹ 60,72,000

Actual factory overheads incurred is as follows:

Production Departments		**Service Departments**	
P_1	**P_2**	**S_1**	**S_2**
₹ 25,41,000	₹ 22,44,000	₹ 6,60,000	₹ 5,28,000

You are required to:

(i) COMPUTE the pre-determined overhead rate for department P1 and P2.

(ii) PREPARE a comparative statement reflecting Budgeted cost and Actual cost for production of the Products X and Y during the month of December, 2024.

(iii) CALCULATE the amount of under/ over-absorption of production overheads

Cost Sheet

11. (a) The figures listed below are derived from the Trial Balance of EVS & Co. as of 31st March:

	Dr. (₹)	Cr. (₹)
Opening Inventories:		
Finished Stock	10,96,000	
Raw Materials	19,18,000	
Work-in-Process	27,40,000	
Office Appliances	2,38,380	
Plant & Machinery	63,08,850	
Building	27,40,000	
Sales		1,35,21,600
Sales Return and Rebates	1,91,800	
Cash discount allowed on sales	1,17,820	
Materials Purchased	43,84,000	
Freight incurred on Materials	2,19,200	
Purchase Returns		65,760
Direct employee cost	21,92,000	
Indirect employee cost	2,46,600	
Drawing and Designing cost	1,37,000	
Repairs and maintenance of factory	1,91,800	
Heat, Light and Power expenses	8,90,500	
Pollution Control Expenses	2,56,190	
Sales Commission	4,60,320	
Sales Promotion	3,08,250	
Distribution Deptt. - Salaries and Expenses	2,46,600	
Office - Salaries and Expenses	1,17,820	
Packing Cost to make the product marketable	3,15,100	

Printing and Stationery expenses	89,050	
Bank Charges paid	8,220	

Further details are available as follows:

(i)	**Closing Inventories:**	
	Finished Goods	₹ 15,75,500
	Raw Materials	₹ 24,66,000
	Work-in-Process	₹ 26,30,400
(ii)	Outstanding direct employee cost	₹ 1,09,600
(iii)	Depreciation to be provided on:	
	Office Appliances	15%
	Plant and Machinery	40%
	Buildings	10%
(iv)	70% of the Heat, Light and Power expenses is related to the Factory and the remaining 30% is equally shared between the Office and Selling Department.	
	Depreciation on Buildings is to be distributed at a similar percentage between the Factory, Office and the Selling Department as that of the Heat, Light and Power.	

With the help of the above information, you are required to PREPARE a condensed Profit and Loss Statement of EVS & Co. for the year ended 31st March along with the following schedules:

(i) Cost of Sales (showing Prime Cost, Gross Works Cost, Cost of production, Cost of Goods Sold and Cost of Sales)

(ii) Selling and Distribution Expenses.

(iii) Administration Expenses

Cost Accounting Systems

12. Amy Ltd. uses a batch costing system that is fully integrated with its financial accounts. Balances at the beginning of the period is provided below:

Particulars	(₹)
Stores Ledger Control Account	4,83,250
Work-in-Process Control Account	3,86,600
Finished Goods Control Account	6,76,550

Other information is as under:

Particulars	
Materials purchased during the period	14,49,750
Materials issued to production	5,79,900
Total wages paid (1/6th being indirect)	5,79,900
Direct wages charged to batches	3,86,600
Payment for non-productive time of direct workers	1/5th of direct wages paid
Production Overheads incurred	2,31,960
Sales	19,33,000
Cost of Finished Goods Sold	15,46,400
Cost of Goods completed and transferred into finished goods during the period	12,56,450
Physical value of Work-in-Process at the end of the period	7,73,200
Production overhead absorption rate	130% of direct wages charged to Work-in-Process

You are required to PREPARE the following accounts:

(i) Stores Ledger Control Account.

(ii) Wages Control Account.

(iii) Production Overhead Control Account.

(iv) Work-in-Process Control Account.

(v) Finished Goods Control Account.

(vi) Costing Profit and Loss Account.

Job and Batch Costing

13. X Ltd. provides the following cost details for calculating the selling price of Job No. X2X:

Particulars	**Per unit (₹)**
Materials	1,330
Direct wages 18 hours @ ₹ 47.50 (Deptt. X: 8 hours; Deptt. Y: 6 hours; Deptt. Z: 4 hours)	855
Chargeable expenses	95
	2,280
Add: 1/3rd for expenses cost	760
	3,040

Analysis of the Profit/Loss Account

(for the current financial year)

			(₹)		**(₹)**
Materials used			28,50,000	Sales	47,50,000
Direct wages:					
	Deptt. X	1,90,000			
	Deptt. Y	2,28,000			
	Deptt. Z	1,52,000	5,70,000		
Special stores items			76,000		
Overheads:					
	Deptt. X	95,000			
	Deptt. Y	1,71,000			
	Deptt. Z	38,000	3,04,000		
Works cost			38,00,000		
Gross profit c/d			9,50,000		
			47,50,000		47,50,000
Selling expenses			3,80,000	Gross profit b/d	9,50,000

Net profit			5,70,000 9,50,000		9,50,000

It is also noted that average hourly rates for the three Departments X, Y and Z are similar.

You are required to prepare a job cost sheet to DETERMINE the selling price by calculating the entire revised cost using above figures as the base and adding 20% to the total cost.

Process & Operation Costing

14. Following data are available for a product for the month of July, 2024:

Particulars	**Process- I (₹)**	**Process- II (₹)**
Opening work-in- progress	Nil	Nil
Costs incurred during the month:		
- Direct materials	6,00,000	
- Labour	1,20,000	1,60,000
- Factory overheads	2,40,000	2,00,000
Units of production:		
Received in process	40,000	36,000
Completed and transferred	36,000	32,000
Closing work-in-progress	2,000	?
Normal loss in process	2,000	1,500

Production remaining in process has to be valued as follows:

Materials 100% Labour 50% Overheads 50%

There has been no abnormal loss in Process- II.

The company follows weighted average method for valuing inventory.

PREPARE Process Accounts after working out the missing figures and with detailed workings.

Joint Products and By products

15. JB Ltd. manufactures three chemical products, XR, YS, and ZT. It processes input material in common plant facility to generate two

intermediate products, X and Y, in 4:1 proportion after a normal loss of $1/12^{th}$ of the input material. There is no market for these two intermediate products. Thus, X is processed further through separate finishing process R to yield the product XR and Y is converted into product YS by a process S. The S finishing process also produces a waste material, Z, which has no market value. Thus, the company converts Z, after additional processing through process T, into a saleable by-product, ZT.

11,40,000 kg of the common input material are processed each month in common plant facility. And after the separate finishing processes (all losses are normal losses), proportions of XR, YS and ZT emerge as follows:

Product	Quantity (kg)	Market price per kg (₹)
XR	7,60,000	38.80
YS	1,90,000	72.00
ZT	19,000	24.00

The material and processing costs are as follows:

Particulars	Common plant facility (₹)	Separate finishing processes		
		R (₹)	S (₹)	T (₹)
Direct material	97,28,000	33,44,000	4,56,000	30,400
Direct labour	45,60,000	68,40,000	27,36,000	1,67,200
Factory overhead	24,32,000	22,80,000	9,12,000	1,06,400
Total	1,67,20,000	1,24,64,000	41,04,000	3,04,000

You are required to CALCULATE the cost per unit and total operating profit/loss attributed to both the products XR and YS considering all joint costs are allocated based on net realisable value method.

Service Costing

16. Mr. Intell newly sets up a Home Stay in Hanle, Ladakh offering two types of room, single and double. The expected occupancy percentage for the rooms is provided below:

Type of rooms	Number	Occupancy percentage
Single	20	100%
Double	10	80%

The details of the expenses as forecasted are provided below:

Particulars	(₹)
Staff salaries	30,20,000
Food and beverage costs	20,16,000
Lighting and power	8,60,000
Repairs and renovation	4,94,000
Laundry charges	3,22,000
Building rent	14,40,000
Miscellaneous expenses	6,12,000

Room attendants to be paid @ ₹ 125 per room day.

Mr. Intell knowing about the power cut during nights at Hanle to preserve the natural light for astronomical research, sets up emergency power backup, being charged from the customers separately @ ₹ 200 per room, if requested.

The rent of the double room is to be fixed at 1.5 times the single room.

In the month of June, Mr. Matrix along with his family visited Hanle Dark Sky Reserve to experience the breathtaking view of the stars. For one night stay at the place, he approached Mr. Intell to book a double bedroom and also requested for emergency power backup at night.

You are required to CALCULATE the rent to be charged from Mr. Matrix, considering the profit @ 25% on total taking.

(Assume 360 days in a year for calculation purpose.)

Standard Costing

17. COMPUTE the missing data indicated by the question marks from the following:

Particulars	A	B
Standard Price/ unit	₹ 12	₹ 15
Actual Price/ unit	₹ 15	₹ 20
Standard Input (kgs.)	50	?
Actual Input (kgs.)	?	70
Material Price Variance	?	?
Material Usage Variance	?	₹ 300 Adverse
Material Cost Variance	?	?

Material mix variance for both products together was ₹ 45 Adverse.

Marginal Costing

18. Gourmet Food Products is a new entrant in the market for chocolates. It has introduced a new product—Sweetee. This is a small rectangular chocolate bar. The bars are wrapped in aluminium foil and packed in attractive cartons containing 50 bars. A carton, is therefore, considered the basic sales unit. Although management had made detailed estimates of costs and volumes prior to undertaking this venture, new projections based on actual cost experience are now required.

Income Statements for the last two quarters are each thought to be representative of the costs and productive efficiency we can expected in the next few quarter. There were virtually no inventories on hand at the end of each quarter. The income statements reveal the following:–

	First Quarter (₹)	Second Quarter (₹)
Sales :		
50,000 × ₹ 24	12,00,000	—
70,000 × ₹ 24	—	16,80,000
Less: Cost of Goods Sold	7,00,000	8,80,000

Gross Margin	5,00,000	8,00,000
Less: Selling and Administration	6,50,000	6,90,000
Net Income / (Loss) before Taxes	(1,50,000)	1,10,000
Less: Tax	(60,000)	44,000
Net Income / (Loss)	(90,000)	66,000

The firm's overall marginal and average income tax rate is 40%. This 40% figure has been used to estimate the tax liability arising from the chocolate operations.

REQUIRED:

(a) Management would like to know the breakeven point in terms of quarterly carton sales for the chocolates.

(b) Management estimates that there is an investment of ₹ 30,00,000 in this product line. What quarterly carton sales and total revenue are required in each quarter to earn an after tax return of 20% per annum on investment?

(c) The firm's marketing people predict that if the selling price is reduced by ₹ 1.50 per carton (₹ 0.03 off per chocolate bar) and a ₹ 1,50,000 advertising campaign among school children is mounted, sales will increase by 20% over the second quarter sales. Should the plan be implemented?

Budgets and budgetary control

19. The budgets for activity and cost of PQR Ltd. for the first three quarters of operation are shown below:

Period Covered	Budgets Quarters I – III		
	Q – I	**Q – II**	**Q – III**
Months	**1 – 3**	**4 – 6**	**7 – 9**
	('000)	**('000)**	**('000)**
Activity :			
Sales (Units)	9	17	15
Production (Units)	10	20	15

Costs (₹) :			
Direct Material			
A	60	120	90
B	50	100	75
Production Labour	180	285	230
Manufacturing Overheads Excluding Depreciation	90	120	105
Depreciation of Production Machinery	20	20	20
Administration Expenses	25	25	25
Selling & Distribution Expenses	38	54	50

The figures shown above represent the costs structure of PQR Ltd., which have the following major features:

(i) Fixed element of any cost is completely independent of activity levels.

(ii) Any variable element of each cost displays a linear relationship with activity level, except that the variable labour cost become 50% higher for activity in excess of 19,000 units per quarter due to the necessity for overtime working.

(iii) The variable element of selling and distribution expenses is a function of sales. All other costs with a variable element are a function of production volume.

Activity for each quarter is spread evenly throughout that quarter.

In Quarter IV Production level will be set equal to sales level. Production and sales in this quarter is expected to range between 15,000 units and 21,000 units. The most likely volume is 18,000 units. In month 9 it will be possible to accurately estimate the sales for Quarter IV.

Cost structure will remain the same as in Quarters I to III except the following:

(i) Labour cost will rise by 12½%.

(ii) Variable labour input per unit of output will decrease, due to learning curve effect, such that 80% of the previous labour input per unit of output will be required in Quarter IV. The threshold for overtime working remains at 19,000 units per quarter.

(iii) Fixed factory overheads and the fixed element of selling and distribution costs will each rise by 20% (The variable element of selling and distribution costs will be unaltered.)

Required

(i) PREPARE a Statement to show, under each cost classification given in the budgets, the variable cost per unit and fixed costs which will be effective in Quarter IV.

(ii) PREPARE a flexible budget of production costs for the Quarter IV.

Miscellaneous

20. (a) DEFINE Product costs. Describe three different purposes for computing product costs.

(b) WHAT do you understand by Operating Costs? DESCRIBE its essential features and state where it can be usefully implemented?

(c) How apportionment of joint costs upto the point of separation amongst the joint products using market value at the point of separation and net realizable value method is done? DISCUSS.

(d) EXPLAIN:

(i) Pre-production Costs

(ii) Research and Development Costs

(iii) Training Costs